Focused blind deconvolution of interferometric Green's functions

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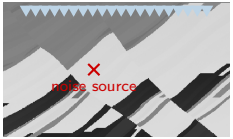
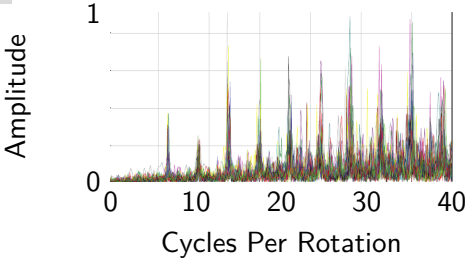
Can We Use Noise-Source Records for Imaging?

- Noise source:
  - is uncontrollable and continuously inputs energy.
  - likely is heavily *correlated* (not white spectrum) in time.
- Imaging:
  - from raw records is impossible because of unknown noise signature.
  - requires the subsurface Green's function that is uncontaminated by noise.

Marmousi Model



Drill Noise Spectrum
(Gradl et al. [2012])



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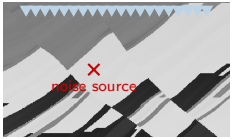

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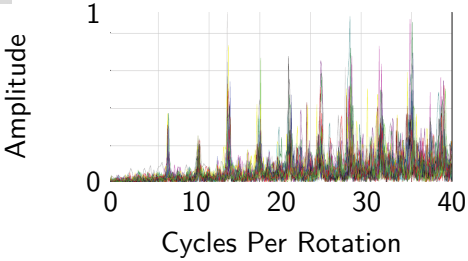
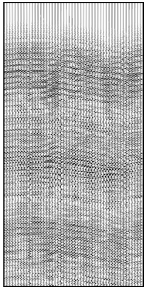


Can We Use Noise-Source Records for Imaging?

Raw Records

t=0s



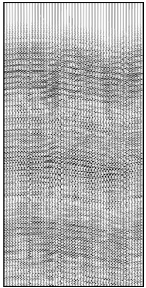
t=7.5s

Marmousi Model

...just using a cross-correlation is not sufficient

Raw Records

t=0s



t=7.5s

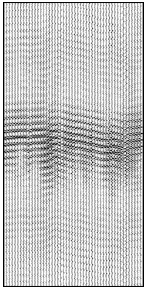
Marmousi Model

Cross-correlation
With Pilot Record

τ=-2s

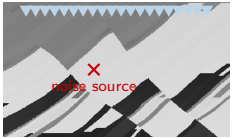
τ=0s

τ=2s



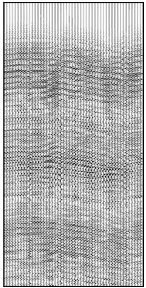
Focused Blind Deconvolution (FBD)

Marmousi Model



Raw Records

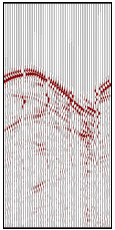
t=0s



t=7.5s

Green's Function

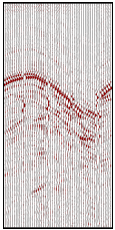
t=0s



t=1s

FBD of Noise

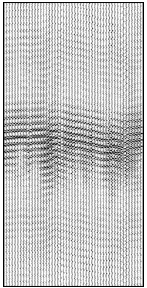
t=0s



t=1s

Cross-correlation
With Pilot Record

τ=-2s



τ=0s

τ=2s

Overview

1. Multichannel Blind Deconvolution
   - Non-uniqueness
2. Two Focusing Constraints
   - Maximally White
   - Maximally Front-loaded
3. Focused Blind Deconvolution
4. Numerical Experiment
5. Conclusions

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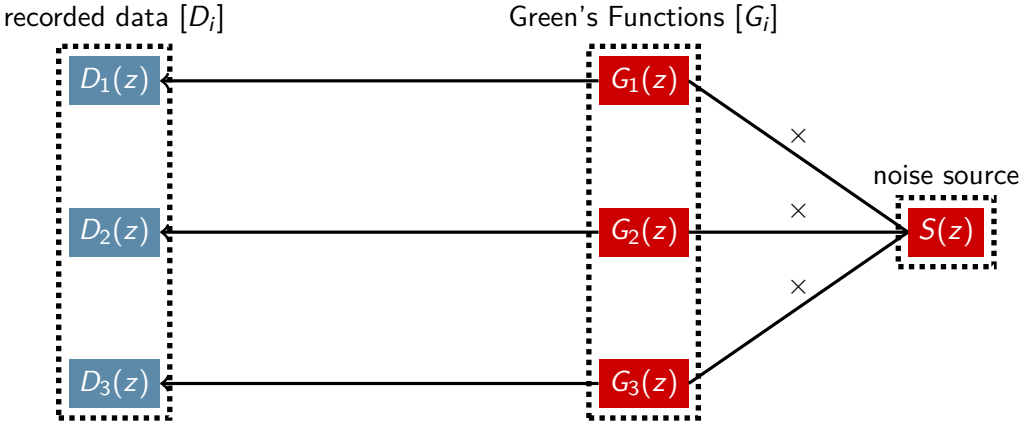
Convolutional Model For Three Channels



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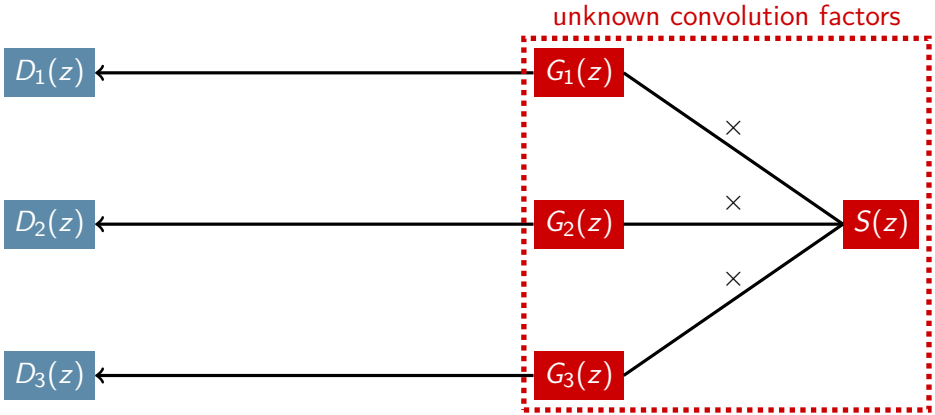
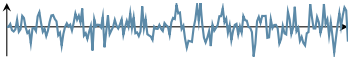
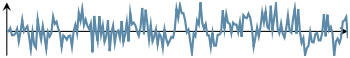


True Solution

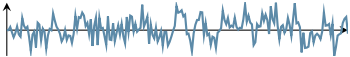
$d_1(t)$



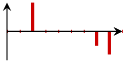
$d_2(t)$



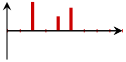
$d_3(t)$



$g_1(t)$

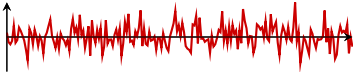


$g_2(t)$



$g_3(t)$

$s(t)$



Least-squares Blind Deconvolution

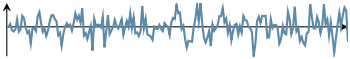
Definition

$$Q(s,[g_i]) = \sum_k \sum_t \{d_k(t) - \{s * g_k\}(t)\}^2 + \alpha \underbrace{\sum_k \sum_\tau t^2 \{g_k \otimes g_k\}^2(\tau)}_{\text{Maximally White}} + \beta \underbrace{\sum_t t^2 g_1(t)^2}_{\text{Maximally Front-loaded}}$$

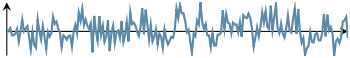
$$(\widehat{s},[\widehat{g}_i]) = \underset{s,[g_i]}{\arg\min} \quad Q$$

Undesired Solution 1: Less Sparse Green's Function

$d_1(t)$



$d_2(t)$



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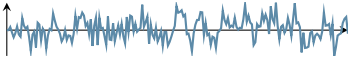

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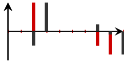
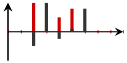
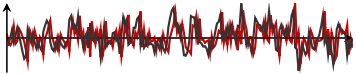


$g_2(t)$



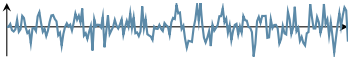
$g_3(t)$

$s(t)$



Undesired Solution 2: Unlikely Source

$d_1(t)$



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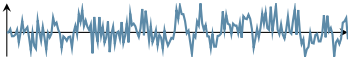

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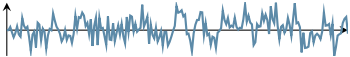

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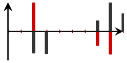

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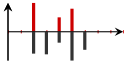

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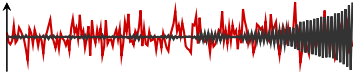

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- Non-uniqueness

2 Two Focusing Constraints
- Maximally White
- Maximally Front-loaded

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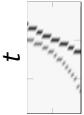
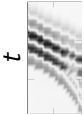
4 Numerical Experiment

5 Conclusions

Green's Function in Seismics: True Vs. Undesired





True

- ⇒ sparse reflections
- ⇒ coprime (see Xu et al. [1995])
- ⇒ auto-correlations maximally focused at zero lag time
- ⇒ minimal variation of phase spectra, i.e., maximally front-loaded

Undesired

- ⇒ less sparse
- ⇒ share common roots due to $\Phi(z)$
- ⇒ not focused due to the common roots
- ⇒ not front-loaded due to the common roots

Standard Constraint (Kazemi & Sacchi [2014], Nose-Filho et al. [2015], Liu et al. [2016]):

➔ minimize ℓ_1 norm — does it fully resolve the indeterminacy?!

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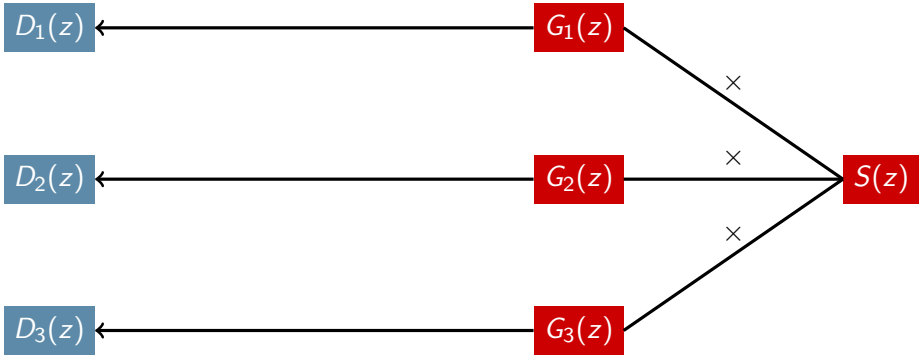
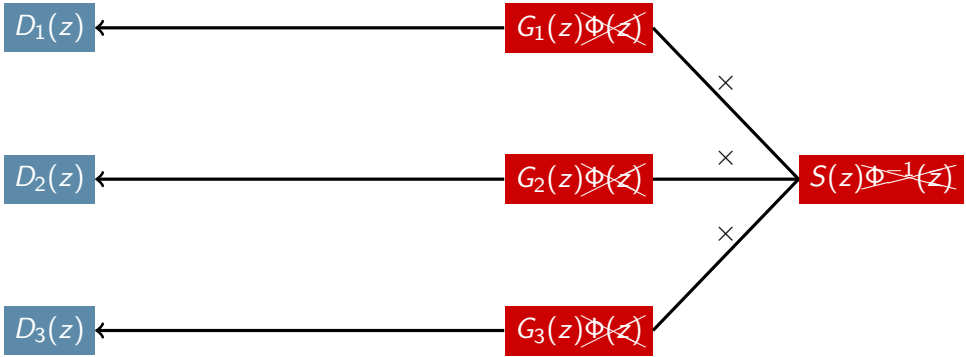


Undesired Green's Function: $\Phi(z) \neq 1$ Is Inserted

True Green's Function: No Common Roots



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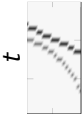

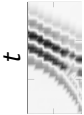

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True Solution: Auto-correlated Green's Function

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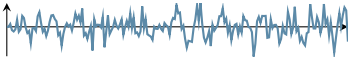
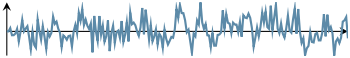
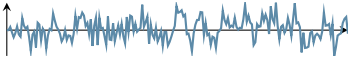


$d_2(t)$



$d_3(t)$



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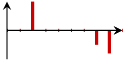
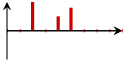
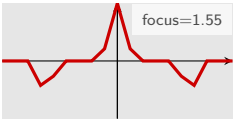


$g_2(t)$

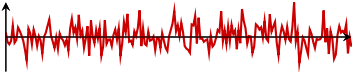


$g_3(t)$

$\{g_1 \otimes g_1\}(\tau)$

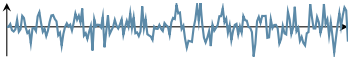


$s(t)$

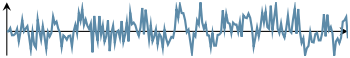


Undesired Solution 1: Auto-correlated Green's Function

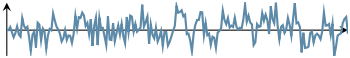
$d_1(t)$



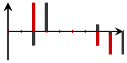
$d_2(t)$



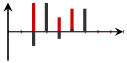
$d_3(t)$



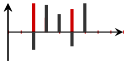
$g_1(t)$



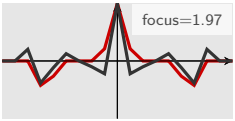
$g_2(t)$



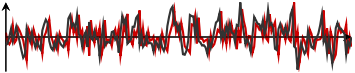
$g_3(t)$



$\{g_1 \otimes g_1\}(\tau)$

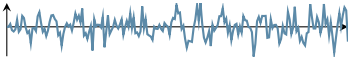


$s(t)$

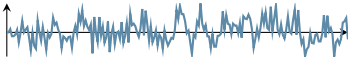


Undesired Solution 2: Auto-correlated Green's Function

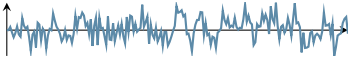
$d_1(t)$



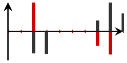
$d_2(t)$



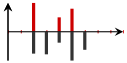
$d_3(t)$



$g_1(t)$



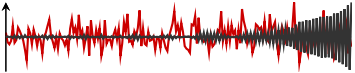
$g_2(t)$



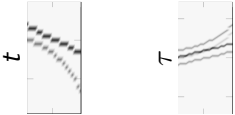
$g_3(t)$

$\{g_1 \otimes g_1\}(\tau)$

$s(t)$



Green's Function in Seismics: True Vs. Undesired



True

- ⇒ sparse reflections
- ⇒ coprime (see Xu et al. [1995])
- ⇒ auto-correlations maximally focused at zero lag time
- ⇒ minimal variation of phase spectra, i.e., maximally front-loaded

Undesired

- ⇒ less sparse
- ⇒ share common roots due to $\Phi(z)$
- ⇒ not focused due to the common roots
- ⇒ not front-loaded due to the common roots

Least-squares Blind Deconvolution

Definition

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$$(\widehat{s},[\widehat{g}_i]) = \underset{s,[g_i]}{\arg\min} \quad Q$$

Adding Focusing Constraint

Definition

$$Q(s,[g_i]) = \sum_k \sum_t \{d_k(t) - \{s * g_k\}(t)\}^2 + \alpha \underbrace{\sum_k \sum_\tau t^2 \{g_k \otimes g_k\}^2(\tau)}_{\text{Maximally White}} + \beta \underbrace{\sum_t t^2 g_1(t)^2}_{\text{Maximally Front-loaded}}$$

$$(\widehat{s},[\widehat{g}_i]) = \underset{s,[g_i]}{\arg\min} \quad Q$$

Next Subsection

1 Multichannel Blind Deconvolution
- Non-uniqueness

2 Two Focusing Constraints
- Maximally White
- Maximally Front-loaded

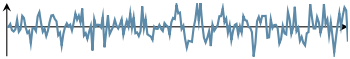
3 Focused Blind Deconvolution

4 Numerical Experiment

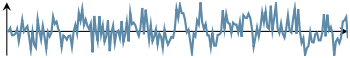
5 Conclusions

True Solution: Phase Spectrum of the Green's Function

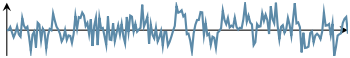
$d_1(t)$



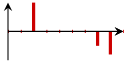
$d_2(t)$



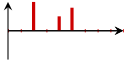
$d_3(t)$



$g_1(t)$

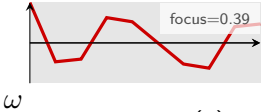


$g_2(t)$

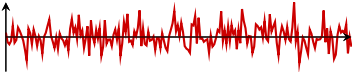


$g_3(t)$

phase spectrum of $g_1(t)$

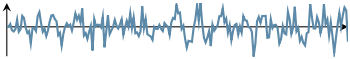


$s(t)$

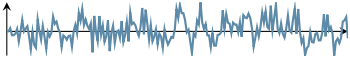


Undesired Solution 1: Phase Spectrum of the Green's Function

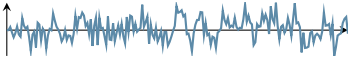
$d_1(t)$



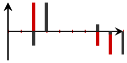
$d_2(t)$



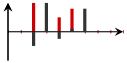
$d_3(t)$



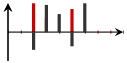
$g_1(t)$



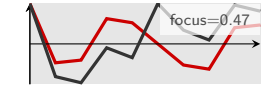
$g_2(t)$



$g_3(t)$

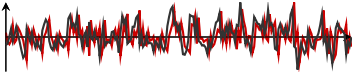


phase spectrum of $g_1(t)$



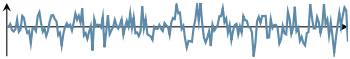
ω

$s(t)$

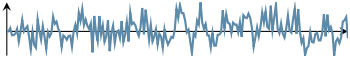


Undesired Solution 2: Phase Spectrum of the Green's Function

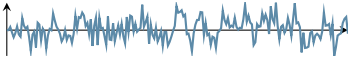
$d_1(t)$



$d_2(t)$



$d_3(t)$



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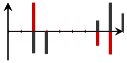
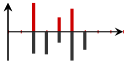
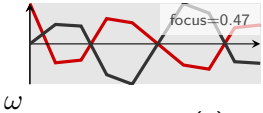


$g_2(t)$



$g_3(t)$

phase spectrum of $g_1(t)$



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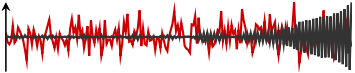

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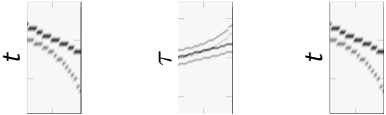

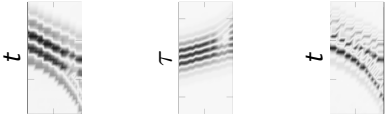

True

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Least-squares Blind Deconvolution

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Next Section

1 Multichannel Blind Deconvolution
- Non-uniqueness

2 Two Focusing Constraints
- Maximally White
- Maximally Front-loaded

3 Focused Blind Deconvolution

4 Numerical Experiment

5 Conclusions

Focused Blind Deconvolution

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$$Q(s,[g_i]) = \sum_k \sum_t \{d_k(t) - \{s * g_k\}(t)\}^2 + \alpha \underbrace{\sum_k \sum_\tau t^2 \{g_k \otimes g_k\}^2(\tau)}_{\text{Maximally White}} + \beta \underbrace{\sum_t t^2 g_1(t)^2}_{\text{Maximally Front-loaded}}$$

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Focused Blind Deconvolution (FBD)

Blind Deconvolution

$$[d_i] = [s * g_i] = [\{s * \phi^{-1}\} * \{g_i * \phi\}]$$

Indeterminacy: ϕ

Focused Blind Deconvolution (FBD)

Blind Deconvolution

$[d_i] = [s * g_i] = [\{s * \phi^{-1}\} * \{g_i * \phi\}]$

Indeterminacy: ϕ

Indeterminacy: amplitude spectrum of ϕ

Indeterminacy: phase spectrum of ϕ

Focused Blind Deconvolution (FBD)

Blind Deconvolution
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Indeterminacy: ϕ

Indeterminacy: amplitude spectrum of ϕ

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Constraint:
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Focused Blind Deconvolution (FBD)

Blind Deconvolution
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Indeterminacy: ϕ

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Constraint:
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Indeterminacy: phase spectrum of ϕ

Constraint:
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Focused Blind Deconvolution (FBD)

Blind Deconvolution
$[d_i] = [s * g_i] = [\{s * \phi^{-1}\} * \{g_i * \phi\}]$
Indeterminacy: ϕ

Indeterminacy: amplitude spectrum of ϕ

Constraint:
$[g_i]$ should be maximally *white*

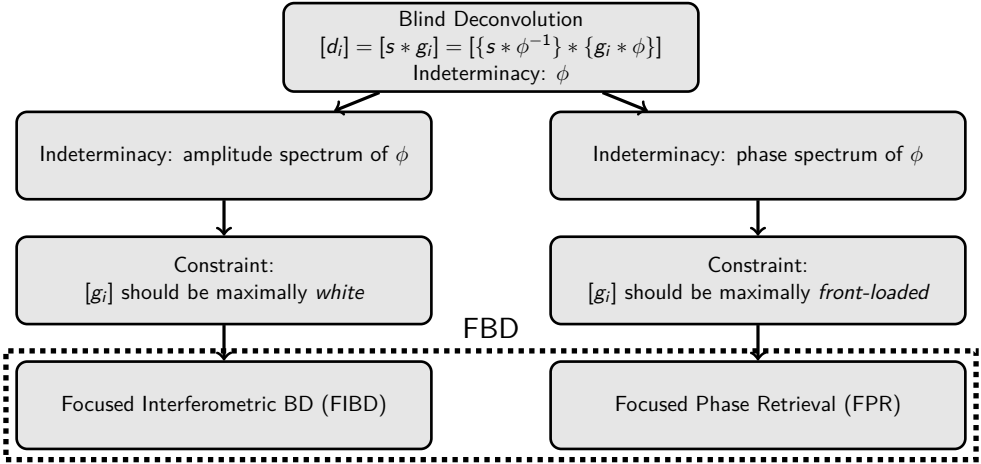
Focused Interferometric BD (FIBD)

Indeterminacy: phase spectrum of ϕ

Constraint:
$[g_i]$ should be maximally *front-loaded*

Focused Phase Retrieval (FPR)

Focused Blind Deconvolution (FBD)



Next Section

1 Multichannel Blind Deconvolution
- Non-uniqueness

2 Two Focusing Constraints
- Maximally White
- Maximally Front-loaded

3 Focused Blind Deconvolution

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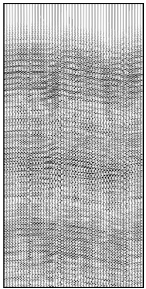
Focused Blind Deconvolution (FBD)

Marmousi Model

Raw Records

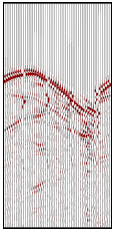
t=0s



t=7.5s

Green's Function

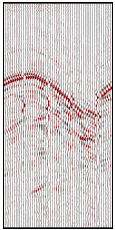
t=0s



t=1s

FBD of Noise

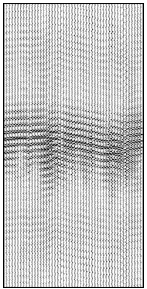
t=0s



t=1s

Cross-correlation
With Pilot Record

τ=-2s



τ=0s

τ=2s

Next Section

1. Multichannel Blind Deconvolution
   - Non-uniqueness
2. Two Focusing Constraints
   - Maximally White
   - Maximally Front-loaded
3. Focused Blind Deconvolution
4. Numerical Experiment
5. Conclusions

Conclusions

Focused Blind Deconvolution:

- ➔ achieves data-driven Green's function retrieval from the multi-channel seismic data of a noisy source.
- ➔ and it doesn't demand either velocity model or unrealistic noisy-source assumptions.

Two Focusing Constraints:

- ➔ resolve the indeterminacy due to the amplitude spectrum of $\phi(t)$, by choosing the Green's functions that are maximally white.
- ➔ resolve the indeterminacy due to the phase spectrum of $\phi(t)$, by choosing the Green's functions that are maximally front-loaded.

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Thank You! Any Questions?

Further Reading

- ➔ Bharadwaj, P., L. Demanet, and A. Fournier, 2018, Focused blind deconvolution: IEEE transactions on signal processing (in review); and on arXiv.org.
- ➔ Xu, G., H. Liu, L. Tong, and T. Kailath, A least-squares approach to blind channel identification, IEEE Transactions on signal processing, vol. 43, no. 12, pp. 2982–2993, 1995.
- ➔ Ayers, G., and J. C. Dainty, 1988, Iterative blind deconvolution method and its applications: Optics letters, 13, 547–549.
- ➔ Robinson, E. A., and S. Treitel, 1980, Geophysical signal analysis: Prentice-Hall Englewood Cliffs, NJ, 263.

- ➔ Download This Presentation: *pawbz.github.io*